

NumStability benchmark: post-hoc ten-task sensitivity report

28 September 2026 | Derived from the sealed Pilot 34/35 fifteen-task composite

Abstract

After viewing the fifteen-task outcomes, five tasks were selected for removal: HI21-EQ2.7, P14-SHIFTED-SUM, P14-ALT-SOFTMAX, P14-SHIFTED-LOG, and P14-SHIFTED-WEIGHTED. No contestant was rerun and no source result was altered. In the remaining ten, L (Mathlib plus NumStability) used 19.7% fewer proof-code lines and 22.5% less contestant-active time than N (Mathlib only), excluding the one-time warm scout. Charging that scout once leaves a 19.9% time gain. Across realized-reuse scores 1, 2, and 3, the selected cohort’s aggregate time gains are -9.1% , $+19.3\%$, and $+31.1\%$. This is a stronger *descriptive* pattern than in the complete corpus. It is not an unbiased or prospective test: both task exclusions and the five prior favorable retained tasks depend on observed outcomes; the reuse rubric was also annotated after the outcomes. Among only the five newly screened tasks still included, L is 7.2% faster but uses 6.4% more proof lines.^[1, 2, 3, 4]

1 What changed, and what did not

This is a subset analysis of the previously published fifteen-task report, *not* a replacement benchmark. The user named five exclusions after inspecting their scores and results on 28 September 2026. The selection rule was not registered before data collection. The original fifteen-task analysis, its negative results, frozen prompts and packets, candidate hashes, audit records, and proof artifacts remain intact. The generated JSON records the exact excluded IDs and input SHA-256 hashes. All fifteen tasks had faithful statements and zero-hole Lean-accepted proofs in both conditions.^[1, 2, 4]

The two conditions, formalization/proof stages, four-submission limits per stage, independent off-clock faithfulness audit, full source access, warm L orientation, three parallel resource-isolated lanes (eight logical CPUs and 24 GiB each), and hardware are unchanged from the primary report. Both formalizers used GPT-6 Sol/high; auditors used GPT-6 Astra/high. No new measurement was made here. “Time” below is summed contestant-active seconds; it excludes audit effort and concurrent campaign makespan. Positive gain means L required less time, fewer net-new tokens, or fewer nonblank proof-code lines.^[1, 2]

Cohort	n	N sec	L sec	Time gain	N lines	L lines
All original tasks	15	10,515	8,978	+14.6%	4,913	4,413
Post-hoc selected	10	6,979	5,410	+22.5%	3,158	2,537
Selected prior favorable	5	3,985	2,631	+34.0%	1,869	1,166
Selected newly screened	5	2,994	2,779	+7.2%	1,289	1,371

Table 1: Summed results from the sealed pairs. Proof-line gains, respectively, are $+10.2\%$, $+19.7\%$, $+37.6\%$, and -6.4% . “Prior favorable” is an outcome-aware retained stratum, not a randomized group.^[2, 4]

2 The ten selected tasks

Task	Reuse score	Origin	Time gain (%)	Proof-line gain (%)
CAST08-FIXED3	3/3	prior	+35.9	+32.0
P14-T1	2/3	new	+24.9	-12.9
HI21-LEM2.2	1/3	new	-47.1	-25.6
CAST08-PROP3.1	3/3	prior	+35.5	+46.4
P14-T2	2/3	new	+14.6	+5.2
RUMP12-THM3.4	3/3	prior	+21.0	+44.4
P14-LOGSUMEXP	3/3	new	+10.5	-26.0
H20-8	3/3	prior	+55.5	+31.3
CAST08-PROP3.2	3/3	prior	+31.6	+25.5
P14-SHIFTED-SOFTMAX	1/3	new	-3.8	-2.5

Table 2: All ten selected tasks in original scheduled order. A positive gain favors L. “Prior” means one of five already known favorable tasks retained in the original corpus; “new” means newly screened for this pilot. [2, 3, 4]

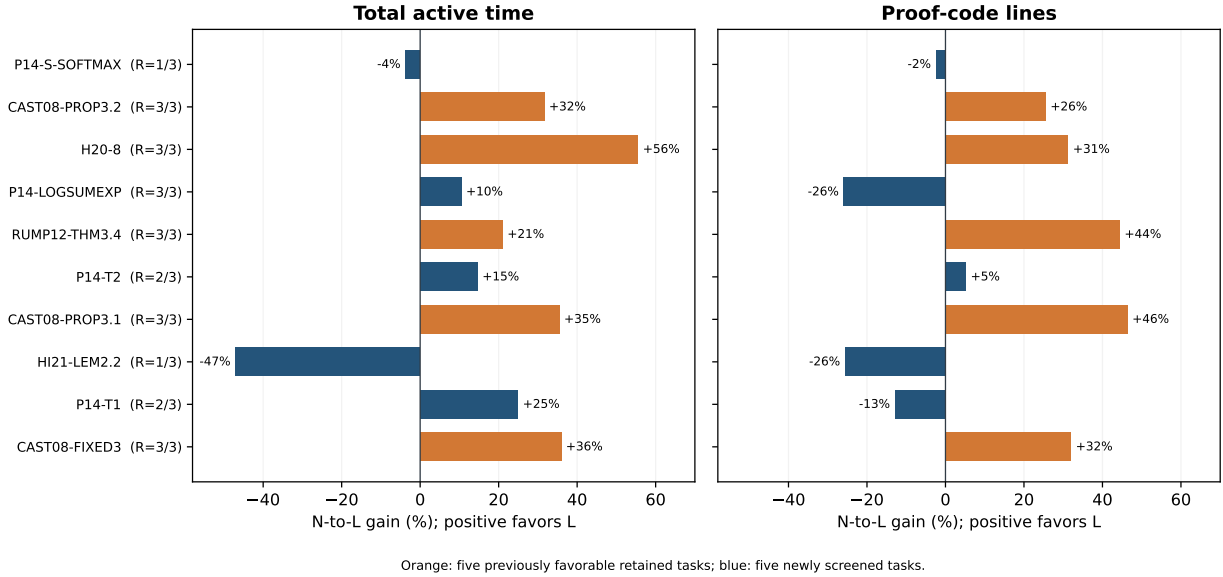


Figure 1: Per-task paired gains. Every task label includes its exact realized reuse score in parentheses. The chart distinguishes prior favorable tasks from newly screened tasks rather than pooling them invisibly. [4]

3 Does the selected cohort support the overlap claim?

The 0–3 score counts *roles* of task-relevant, directly used NumStability declarations in the final L statement or kernel-checked proof: foundation, computation interface, and mathematical analysis. It does not count imports or total declarations and is not an independent measure of potential library coverage. The annotation is post-outcome. The five exclusions remove the only score-0 task, one score-2 task, and three score-3 tasks. The retained score counts are 2, 2, and 6 for scores 1, 2, and 3.^[3, 4]

Reuse score	n	Active-time gain (%)	Proof-line gain (%)	Token gain (%)
1/3	2	-9.1	-4.2	-41.0
2/3	2	+19.3	-1.2	-20.2
3/3	6	+31.1	+31.4	-37.3

Table 3: Aggregate N-to-L gains within each retained score stratum. Positive favors L. There is no score-0 task left in this selected set; score-2 proof lines do not improve. Small, unequal and selected groups preclude a causal trend claim.^[4]

The selected score strata form a monotone sequence in *aggregate active time*: -9.1% , $+19.3\%$, $+31.1\%$. The proof-line sequence is -4.2% , -1.2% , $+31.4\%$; hence a substantial code-size reduction appears only in the highest stratum. This supports the narrower observation that, *in this outcome-selected subset*, broader realized reuse accompanies better aggregate time and proof-code outcomes. It does not show that more reuse caused the gains. Five of the six score-3 tasks are the previously favorable retained tasks. The only new score-3 task, P14-LOGSUMEXP, is faster but has 26.0% *more* L proof lines.^[3, 4]

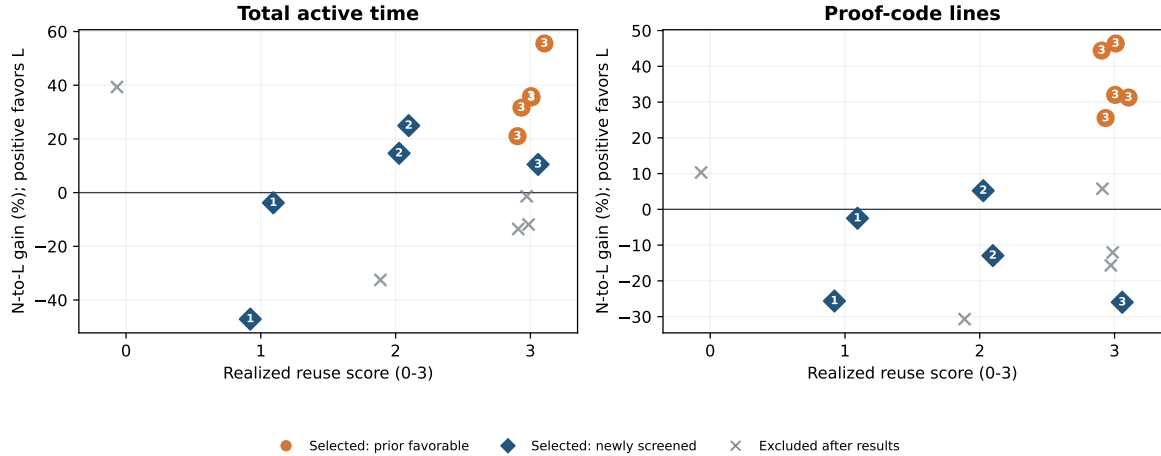


Figure 2: All original fifteen pairs remain visible. Grey crosses mark the five post-hoc exclusions; orange circles and blue diamonds are the ten included pairs. The numeral inside each included point is its reuse score; horizontal jitter only separates integer-score ties. These panels expose the selection behind the apparent score relationship.^[2, 3, 4]

4 Cost decomposition and limits of the claim

For the ten selected tasks, N and L required 1,389 and 1,490 formalization seconds, respectively: L was 7.3% slower before a faithful statement. In the proof stage, they required 5,590 and 3,920 seconds: L was 29.9% faster. The one-time task-neutral L orientation cost another 183 seconds, making the complete ten-task time sums 6,979 (N) versus 5,593 (L), a 19.9% L gain. This is an amortized ten-task sum, not the wall-clock makespan of three parallel lanes.^[2, 1, 4]

The token picture is unfavorable to L. Formalization used 399,748 N versus 980,733 L net-new tokens; proof used 873,571 N versus 734,899 L. Overall, the ten pairs used 1,273,319 N versus 1,715,632 L net-new tokens: 34.7% *more* for L. Charging the warm scout’s 88,468 tokens once raises L to 1,804,100, or 41.7% more than N. Task-time search and compilation are included or excluded according to the frozen controller’s contestant clock; the present filtered tables cannot isolate search as the cause of any one negative result. Conversation traces would be required to establish such a mechanism.^[2, 1, 4]

Task removed after inspection	Score	Time gain (%)	Proof-line gain (%)
P14-ALT-SOFTMAX	3/3	-1.4	-15.7
P14-SHIFTED-SUM	0/3	+39.3	+10.3
P14-SHIFTED-LOG	3/3	-13.5	+5.8
P14-SHIFTED-WEIGHTED	2/3	-32.5	-30.7
HI21-EQ2.7	3/3	-11.9	-12.1

Table 4: The removed observations, retained in the source and shown here to make selection auditable. P14-SHIFTED-SUM was favorable to L despite score zero, whereas several high-score tasks were unfavorable.^[2, 3, 4]

Defensible conclusion. In these ten transparently selected, faithful and fully proved tasks, L reduced total active time and proof-code size; its largest aggregate gains lie in the highest realized-reuse group. That demonstrates useful reuse for a slice of numerical-stability problems. It does *not* establish that lower overlap generally causes loss, that library expansion alone will fix every negative task, or that the reported percentages estimate performance on unseen tasks. The full fifteen-task result remains the primary development evidence. A prospective corpus and predefined score would be needed for a general coverage-benefit claim.

Reproduction and sources

The accompanying README gives the exact three-command rebuild procedure: verify the primary evidence, run this report’s filter script, then compile the TeX source. The script asserts the original 15 identities, the exact five exclusions, and the 5+5 selected origin mix. The derived `generated/summary.json` contains the source SHA-256 hashes, selected IDs, and full-precision aggregates.

References

- [1] NumStability benchmark, complete fifteen-task evidence report, `paper_benchmark/thesis_report/report.tex` and `output/pdf/report.pdf`, branch `codex/numstability-benchmark-thesis-report`, commit `30cda8a55`.
- [2] Hash-checked task-level measurements, `paper_benchmark/thesis_report/generated/task_metrics.csv`; primary verification script `paper_benchmark/thesis_report/verify_release.py`.
- [3] Exact-witness realized-reuse rubric and ledger, `paper_benchmark/thesis_report/reuse_obligations.json` and `paper_benchmark/thesis_report/generated/realized_reuse.csv`.
- [4] This report’s reproducible filter, figures and aggregates, `paper_benchmark/thesis_report/posthoc_10/make_report.py` and `paper_benchmark/thesis_report/posthoc_10/generated/summary.json`.